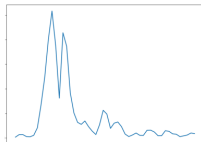
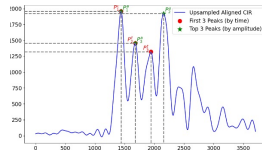


CIR Processing & Localization Running on Smartphone

IMU-based Cross-Seat
Movement Detection



CIR Acquisition from UWB
ranging



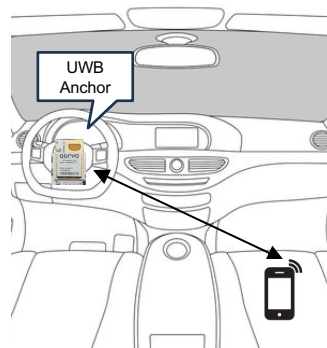
Peak Identification and
Tracking

Peak-based CIR Feature
Extraction

Lightweight ML Classifier

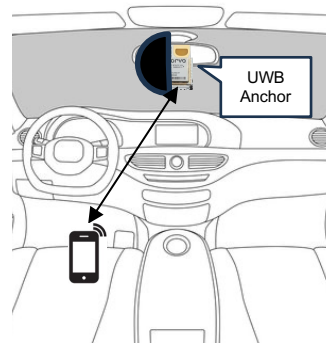
Driver or Passenger Seat?

Steering Wheel Column Mount



Asymmetric UWB anchor placement

Rearview Mirror Mount w/ half-cylindrical metal shield



Shielded UWB anchor